

Internship Training report

On

“DEVOPS & AWS CLOUD”

Submitted in partial fulfillment of the requirements for the award of Degree of
Masters of Computer Application

IN

DEPARTMENT OF COMPUTER APPLICATIONS

SESSION - 2024-25

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1. Acknowledgement

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I am grateful to Engineering College Bikaner and Bikaner Technical University for providing the infrastructure and academic environment necessary for completing this industrial project.

Special thanks to my family and friends for their unwavering support and motivation during the course of this project.

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(Signature)

2. Candidate's Declaration

I, Kamar Raza, hereby declare that the Industrial Project Report entitled "E-pharmacy Prescription" submitted to the Department of Computer Application, Engineering College Bikaner, affiliated to Bikaner Technical University (BTU), Rajasthan, in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications, is a record of original work carried out by me under the guidance of Dr. Abdul Jabbar Khilji MCA Department.

I further declare that this report has not been submitted to any other university or institution for the award of any degree or diploma.

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Engineering College Bikaner

Submitted To

Centre for Cloud & DevOps Training Society, Bikaner

Internship Duration

19 January 2026 to 05 March 2026

CERTIFICATE

This is to certify that Mr. Kamar Raza, student of MCA from Engineering College Bikaner, has successfully completed industrial training cum internship on “**DevOps & AWS Cloud**” at Centre for Cloud & DevOps Training Society, Bikaner from **19.01.2026 to 05.03.2026**. During the training period, he worked on various DevOps and cloud-based technologies and completed all assigned practical tasks and project activities successfully.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Centre for Cloud & DevOps Training Society, Bikaner for providing me the opportunity to complete my internship training on DevOps and AWS Cloud. The organization provided excellent practical exposure and industrial-level learning experience.

I am thankful to the Director, mentors, and trainers for their valuable guidance, support, and encouragement throughout the internship period. Their guidance helped me understand modern cloud infrastructure, deployment automation, and DevOps practices.

I would also like to thank my college faculty members and my family for their continuous motivation and support. The internship helped me improve my technical knowledge,

communication skills, teamwork abilities, and confidence.

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1. INTRODUCTION

Industrial training is an important part of technical education because it helps students gain practical exposure and understand real industry environments. Classroom learning provides theoretical knowledge, while industrial training helps in understanding practical implementation and real-world workflows.

During this internship on DevOps & AWS Cloud, practical knowledge about cloud computing, deployment automation, Linux administration, Docker, Kubernetes, GitHub, and CI/CD pipelines was gained. The internship provided exposure to industry-level tools and technologies used for software deployment and cloud infrastructure management.

The internship also improved problem-solving abilities, teamwork skills, communication skills, and technical understanding. Through practical implementation and project-based learning, the gap between academic learning and industrial requirements was reduced.

Modern organizations are rapidly shifting towards cloud computing and automation technologies. DevOps practices are becoming essential for faster software delivery, improved collaboration, and infrastructure management. This internship provided understanding of modern deployment strategies and cloud-based environments.

2. ABOUT THE ORGANIZATION

Centre for Cloud & DevOps Training Society, Bikaner is a professional training organization that provides industrial training and certification programs in cloud computing, DevOps, Linux administration, networking, and software development.

The organization focuses on practical learning and project-based implementation. Students are provided with hands-on sessions and real-time examples to improve technical understanding and industry readiness.

The training organization provides exposure to technologies such as:

- Linux Administration
- Docker
- Kubernetes
- AWS Cloud
- GitHub

- Jenkins
- CI/CD Tools
- Monitoring Tools

The organization also promotes open-source technologies and practical learning approaches that help students prepare for industrial careers.

3. OBJECTIVES OF INTERNSHIP

The main objectives of the internship were:

- To understand DevOps culture and workflow
- To learn AWS Cloud infrastructure and deployment
- To gain practical knowledge of Linux server administration
- To understand Docker and containerization concepts
- To learn Kubernetes orchestration
- To understand CI/CD automation workflows
- To improve troubleshooting and problem-solving skills
- To gain project implementation experience
- To improve communication and teamwork skills

The internship also aimed to provide industrial exposure and practical understanding of modern cloud technologies.

4. INTRODUCTION TO DEVOPS

DevOps is a combination of **Development** and **Operations**. It is a modern software development methodology that focuses on collaboration, automation, monitoring, and continuous delivery.

In traditional software development, developers and operations teams worked separately, which often caused delays and communication gaps. DevOps removes these barriers and improves collaboration between teams.

Major Components of DevOps

- Continuous Integration (CI)
- Continuous Delivery (CD)

- Infrastructure as Code (IaC)
- Monitoring and Logging
- Automation
- Collaboration and Feedback

Advantages of DevOps

- Faster deployment of applications
- Reduced deployment failures
- Improved software quality
- Better collaboration between teams
- Continuous monitoring and feedback
- Faster issue resolution
- Increased productivity

DevOps Lifecycle

1. Planning
2. Development
3. Building
4. Testing
5. Deployment
6. Monitoring
7. Feedback

DevOps helps organizations deliver applications quickly, efficiently, and with fewer errors.

5. AWS CLOUD COMPUTING

Amazon Web Services (AWS) is one of the leading cloud computing platforms used worldwide. It provides services related to computing, storage, networking, security, databases, monitoring, and deployment.

AWS allows organizations to host applications without maintaining physical infrastructure. It provides scalability, reliability, flexibility, and cost optimization.

Features of AWS

- High scalability
- Global infrastructure
- Data security
- Pay-as-you-use pricing

- High availability
- Easy deployment

Important AWS Services

Service	Purpose
EC2	Virtual server hosting
S3	Cloud storage
IAM	User and permission management
VPC	Networking and isolation
CloudWatch	Monitoring and logging

AWS is widely used because it allows organizations to deploy applications globally with high performance and security.

6. LINUX ADMINISTRATION

Linux is an open-source operating system widely used in cloud servers and enterprise environments. Linux provides stability, security, and flexibility, making it popular for cloud infrastructure.

During the internship, Linux administration concepts and commands were practiced.

Topics Covered

- File and directory management
- User and group management
- File permissions
- Networking commands
- Package management
- Shell scripting
- Process management

Important Linux Commands

Command	Purpose
ls	List files and folders
pwd	Display current directory
mkdir	Create directory
chmod	Change file permissions
grep	Search text
top	Monitor processes
ps	Display running processes
ping	Check connectivity

Linux administration improved understanding of server environments and deployment infrastructure.

7. GIT & GITHUB

Git is a distributed version control system used for tracking changes in source code. GitHub is a cloud-based platform used for repository hosting and collaboration.

Version control systems are important because they help developers manage code changes efficiently and maintain project versions.

Git Commands Practiced

- git init
- git clone
- git add
- git commit
- git push
- git pull
- git branch
- git merge

Advantages of GitHub

- Code collaboration
- Version management
- Backup and recovery
- Team development
- Repository hosting

GitHub also helps in project collaboration, issue tracking, and deployment workflows.

8. DOCKER & CONTAINERIZATION

Docker is a containerization platform used to package applications along with their dependencies into containers. Containers provide consistency across development, testing, and production environments.

Topics Covered

- Docker Images
- Docker Containers
- Dockerfile
- Docker Compose
- Volume Management
- Container Networking

Advantages of Docker

- Lightweight virtualization
- Faster deployment
- Easy portability
- Better scalability
- Efficient resource utilization

Practical sessions included creating Docker containers, building images, and deploying web applications.

9. KUBERNETES

Kubernetes is an orchestration platform used to manage containerized applications. It automates deployment, scaling, and management of containers.

Concepts Covered

- Pods
- Deployments
- Services
- Replica Sets
- Scaling
- Cluster Architecture

Benefits of Kubernetes

- Automatic scaling
- High availability
- Efficient resource management
- Self-healing capability
- Load balancing

Kubernetes is widely used for managing large-scale cloud-native applications.

10. CI/CD PIPELINE

CI/CD stands for Continuous Integration and Continuous Delivery. It automates software build, testing, and deployment processes.

Continuous Integration (CI)

CI automatically integrates code changes into repositories and performs automated testing.

Continuous Delivery (CD)

CD automates deployment processes and ensures faster software delivery.

Advantages of CI/CD

- Faster software delivery
- Reduced deployment errors
- Automated testing
- Improved software quality

- Better collaboration

CI/CD pipelines improve deployment efficiency and reduce manual work.

11. AWS SERVICES USED

The following AWS services were explored during the internship:

AWS Service	Description
EC2	Used for creating virtual servers
S3	Used for object storage
IAM	Used for managing users and permissions
VPC	Used for cloud networking
CloudWatch	Used for monitoring and logging

These services provided practical exposure to cloud infrastructure and deployment workflows.

12. MONITORING & LOGGING

Monitoring tools are important for tracking server health and application performance. Logging helps identify errors and troubleshoot issues quickly.

Monitoring Activities Performed

- Checking CPU usage
- Monitoring memory utilization
- Tracking server performance
- Viewing application logs

Benefits of Monitoring

- Faster issue detection

- Better infrastructure visibility
- Improved system performance
- Efficient troubleshooting

Monitoring tools improve reliability and performance of cloud applications.

13. SECURITY IN CLOUD COMPUTING

Cloud security is important for protecting applications, users, and data. Security practices help prevent unauthorized access and cyber threats.

Security Measures Learned

- IAM user policies
- Access control
- Security groups
- Password management
- Network security

Importance of Cloud Security

- Protects sensitive data
- Prevents unauthorized access
- Improves infrastructure safety
- Maintains application reliability

Security is an important part of modern cloud infrastructure management.

14. WEEK WISE INTERNSHIP REPORT

Week 1 – Introduction to DevOps & Linux

During the first week, introduction to DevOps and Linux operating system was provided. Linux installation, file system structure, basic commands, and server administration concepts were practiced. Networking basics and Linux permissions were also studied.

Hands-on practical sessions helped in understanding Linux administration and command-line operations.

Week 2 – Git & GitHub

In the second week, Git and GitHub were introduced. Repository creation, version control concepts, branch management, and collaboration workflows were practiced.

Practical exercises included pushing source code to GitHub repositories and managing project versions.

Week 3 – Docker & Containerization

The third week focused on Docker and containerization concepts. Docker installation, image creation, Dockerfile writing, and container deployment were practiced.

Docker Compose and container networking concepts were also explored through practical implementation.

Week 4 – Kubernetes

The fourth week focused on Kubernetes orchestration. Concepts such as pods, deployments, replica sets, and services were studied.

Practical sessions included deploying applications on Kubernetes clusters and scaling containers.

Week 5 – AWS Cloud Services

During the fifth week, AWS Cloud services were explored. EC2 instance creation, IAM users, S3 bucket management, VPC networking, and security groups were configured.

Cloud deployment and server hosting activities provided practical understanding of AWS infrastructure.

Week 6 – CI/CD & Final Project

The final week focused on CI/CD pipelines, automation, monitoring, troubleshooting, and final project implementation.

The internship concluded with deployment tasks and project-based practical implementation.

15. PROJECT WORK

During the internship, a cloud-based deployment project was performed. The project involved deploying a web application using Docker containers on an AWS EC2 instance.

Project Workflow

1. Creating AWS EC2 instance
2. Installing Docker and dependencies
3. Cloning project repository from GitHub
4. Creating Docker image using Dockerfile
5. Running containers using Docker Compose
6. Monitoring application logs
7. Testing deployment

This project provided practical understanding of DevOps workflow and cloud deployment strategies.

16. CHALLENGES FACED

During the internship, the following challenges were faced:

- Understanding cloud networking concepts
- Managing Linux permissions and users
- Debugging Docker container issues
- Configuring Kubernetes deployments
- Learning automation workflows in CI/CD pipelines

These challenges improved troubleshooting abilities and practical understanding of

deployment environments.

17. SKILLS GAINED

The internship helped in developing the following skills:

- Linux server administration
- AWS Cloud deployment
- Docker and Kubernetes management
- Git and GitHub collaboration
- CI/CD automation
- Problem-solving and troubleshooting
- Communication and teamwork

The training improved both technical and professional skills.

18. CONCLUSION

The internship on DevOps & AWS Cloud provided valuable industrial exposure and practical understanding of cloud computing and automation technologies. The training improved technical skills, confidence, teamwork, and project implementation abilities.

Practical exposure to Linux, Docker, Kubernetes, AWS Cloud, GitHub, and CI/CD pipelines enhanced understanding of modern deployment workflows and cloud infrastructure.

The experience gained during the internship will be beneficial for future academic projects and professional career growth.